

Question

In dichloroethane at -78°C , $[\text{C}_6\text{F}_5\text{Xe}][\text{AsF}_6]$ reacts with trimethylchlorosilane in a molar ratio of 1:3 to yield n products ($n > 3$), among which three products are a certain 1:1-type salt, AsCl_3 , and Cl_2 , in a molar ratio of 1:1:1. What is the correct statement about this reaction?

- A. Chemical equation (with coefficients in their simplest integer ratios) where the sum of the coefficients on the product side is odd and the sum of the coefficients on the reactant side is even
- B. The xenon element is oxidized
- C. There are 5 types of products.
- D. In xenon-containing products, the xenon is in the sp^3 hybridization state.
- E. If more than one of A~D is correct, select this option
- F. If all options A~D are incorrect, please select this option

Answer

Correct Answer: **A**

Detailed Explanation

The 1:1 salt of the reaction product is likely to contain Xe, and $[(C_6F_5)Xe]^+$ still appears as an intact unit. Considering the large volume of the cation in the 1:1 salt, the anion should also be a bulky complex anion, most likely $[AsF_6]^-$.

CHECKPOINT

1 PTS

The cation of the 1:1 salt contains $[(C_6F_5)Xe]^+$, and the anion is $[AsF_6]^-$

Assuming the coefficients of Cl_2 and $AsCl_3$ in the product are 1, the product side contains 2 As atoms. Setting the coefficient of the reactant $[C_6F_5Xe][AsF_6]$ to 2, the coefficient of $SiCl(CH_3)_3$ becomes 6.

It is observed that one As on the product side is no longer bonded to fluorine atoms. Combined with stoichiometric considerations, it can be inferred that the product includes a substance with a Si – F bond and a coefficient of 6, namely $SiF(CH_3)_3$.

CHECKPOINT

1 PTS

The product contains $SiF(CH_3)_3$

Combining the above information, it is clear that the cation of the 1:1 salt is $[(C_6F_5Xe)_2Cl]^+$, with a bonding arrangement of $C_6F_5 - Xe - Cl - Xe - C_6F_5$. According to VSEPR theory, Xe has $(8 + 2)/2 = 5$ electron

pairs, with 3 lone pairs, resulting in a linear geometry and sp^3d hybridization. The $[(C_6F_5)Xe]^+$ unit remains unchanged, and the oxidation state of xenon is unaffected.

CHECKPOINT

0.5 PTS

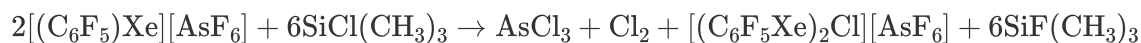
Xe adopts sp^3d hybridization

CHECKPOINT

0.5 PTS

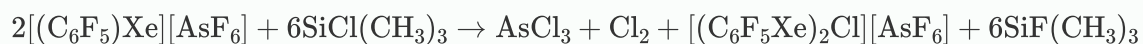
Xe undergoes no redox changes

The reaction equation is:



CHECKPOINT

2 PTS



Only option A is correct. The answer is **A**.